

Article

Cognitive-Emotional Teacher Burnout Syndrome: A Comprehensive Behavioral Data Analysis of Risk Factors and Resilience Patterns During Educational Crisis

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Abstract

Background/Objectives: Teacher burnout represents a complex cognitive-emotional syndrome characterized by the interplay between mental exhaustion and emotional dysregulation, threatening educational sustainability during crisis periods. This study employed comprehensive behavioral data analysis to investigate burnout syndrome patterns among Greek teachers during the COVID-19 educational crisis, aiming to identify risk factors and resilience patterns through multiple analytical approaches that capture the syndrome's multidimensional nature. **Methods:** A cross-sectional study examined primary and secondary school teachers in Western Greece during the autumn of 2021. Stratified random sampling ensured representativeness across school levels, geographic locations, and employment types. Participants completed the Greek-adapted Maslach Burnout Inventory for Educators, which measured emotional exhaustion, depersonalization, and personal accomplishment. Behavioral data analysis integrated traditional statistical methods with advanced pattern recognition techniques, including classification trees for non-linear relationships, association analysis for behavioral patterns, and cluster analysis for profile identification. **Results:** The majority of teachers experienced high stress with inadequate coping capabilities. Classification analysis achieved high accuracy in predicting burnout severity, identifying emotional exhaustion as the primary predictor. Deputy teachers demonstrated severe cognitive-emotional strain compared to permanent colleagues across all dimensions, with dramatically reduced personal accomplishment and minimal resources. Association analysis revealed that combined low support and high workload more than doubled burnout risk. Three distinct profiles emerged: Resilient teachers, characterized by older age and permanent employment; At-Risk teachers, showing early warning signs; and Burned Out teachers, predominantly young and in precarious employment. Remote teaching, exceeding half of the workload, significantly increased strain. Multiple regression confirmed emotional exhaustion as the dominant syndrome predictor. **Conclusions:** Behavioral data analysis revealed complex cognitive-emotional patterns constituting burnout syndrome during educational crisis. Employment precarity emerged as the fundamental vulnerability factor, with young deputy teachers facing dramatically higher syndrome probability compared to supported senior permanent teachers. The syndrome manifests through cascading processes where cognitive overload triggers emotional exhaustion, subsequently reducing personal accomplishment. These findings provide an evidence-based framework for early syndrome identification and targeted interventions addressing both cognitive and emotional dimensions of teacher burnout.



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Keywords: cognitive-emotional burnout syndrome; teacher burnout; behavioral data analysis; educational crisis; employment status; resilience patterns; risk factors; Greece; Maslach Burnout Inventory; COVID-19

1. Introduction

Teacher Burnout Syndrome is a major cognitive-emotional issue that is being confronted by all learning systems around the world, as it also holds a profound effect on teachers' well-being and performance. Burnout was first defined as a state of physical and emotional exhaustion in the 1970s, but it has become a complex burnout syndrome that has three components, including lack of accomplishment, depersonalization, and emotional exhaustion [1,2]. Modern teaching is not just a matter of basic pedagogical skills that need cognitive processing in a limited manner. Rather than that, it is an area that needs cognitive processing related to ever-changing conditions in this field [3–9].

The necessity of addressing teacher burnout syndrome increases as educational systems face unprecedented challenges. Prior to the COVID-19 pandemic, educators encountered increasing cognitive-emotional demands coupled with decreasing resources. Research indicates that approximately 500,000 teachers leave the profession each year in the United States, resulting in economic costs exceeding \$2.2 billion [10–16]. The exodus is driven by various cognitive and emotional stressors, including excessive workload, insufficient compensation, resource scarcity, administrative burdens, and hostile work environments [9,10]. The prevalence of burnout symptoms among educators is alarming, with studies indicating that up to 70% of teachers experience frequent burnout symptoms, while 30% report significant but less frequent manifestations [17–22].

1.1. Theoretical Framework of Teacher Burnout

A basic cognitive-emotional framework for understanding burnout was introduced by Maslach & Jackson [1], including three interdependent components of burnout. Emotional exhaustion indicates a lack of emotional resources, making it psychologically impossible for educators to cope effectively with their duties. This is illustrated through feelings of tiredness, irritability, and a loss of enthusiasm for teaching activities [23–26]. Depersonalization includes feelings of cynicism in reactions to students, parents, and school colleagues through impersonal interactions and loss of emotional engagement. Personal accomplishment is an element of cognitive self-evaluation, and it is illustrated through feelings of ineffectiveness, disappointment in academic achievements, and a lack of self-esteem in one's job performance [27–34].

Job Demands-Resources (JD-R) theory provides a holistic cognitive-emotional model of understanding related to components of burnout [35]. Job demands encompass physical, subjective, social, and organizational demands that require prolonged use of cognitive and emotional faculties, such as classroom management, instruction, assessment, and administrative activities. Job resources refer to those demands that act as a means of accomplishing objectives, diminishing cognitive demands, or enhancing individual improvement, such as autonomy, social support, opportunities for individual improvement, and adequate materials. Burnout occurs when demands continue to exceed job resources, leading to a lack of cognitive and emotional resources [36–44].

Burnout clearly follows a predictable cognitive-emotional process. First, teachers feel a lack of resources to cope with expectations, leading to feelings of stress and anxiety [45]. Then, cynicism is experienced as a cognitive-emotional defense against unmanageable expectations, exhibited through feelings of emotional disconnection between teachers and

students/colleagues. Finally, a lack of efficacy develops as a cognitive accommodation to unachievable expectations, leading to reduced job performance/satisfaction [46,47].

1.2. The COVID-19 Context and Educational Crisis

The sudden need for emergency remote instruction in 2020, due to the pandemic, has produced an unprecedented level of cognitive-emotional workload, greatly exacerbating pre-existing job burnout. Instructors around the globe have been faced with adapting technology skills in a hurry, adapting curricula to remote digital learning delivery, as well as adhering to student engagement and addressing health issues of one's self and one's family within and outside of work when students are undergoing trauma in learning loss as well [48–51].

Research suggests that emergency phases experienced intense burnout escalation as well [52]. Major cognitive-emotional sources of teachers' stress included ensuring students' safety, ensuring equality despite technological disparities, and navigating a new, unexplored arena of online learning [53]. Intensive oscillations between offline, mixed, and online learning in 2020–2021 increased cognitive load because teachers kept modifying their pedagogical approaches, which, in turn, rapidly accelerated teachers' burnout [31,54,55].

1.3. Demographic and Individual Factors in Burnout Syndrome

The association between demographic characteristics and burnout exhibits intricate patterns that require detailed analysis with sophisticated methods. Gender differences in the manifestation of burnout continue to be debated in the literature. Some studies indicate no significant gender differences [56,57], whereas others report increased emotional exhaustion among female teachers [58–61] and greater depersonalization among male teachers. Recent research on the pandemic indicates that traditional gender differences may have lessened as both genders faced unprecedented cognitive-emotional challenges equally [62–65].

Age and experience exhibit more consistent protective effects against cognitive-emotional burnout. Numerous studies indicate that novice teachers are more susceptible to emotional exhaustion and cynicism, and report lower levels of professional efficacy than their more experienced counterparts [66–69]. Teachers younger than 40 are at higher risk of burnout, with symptoms generally diminishing after age 45 [70,71]. This resilience associated with aging likely reflects the accumulation of cognitive coping strategies, realistic expectations, and the consolidation of professional identity [72–75].

The relationship between experience and burnout shows similar patterns, though the findings differ. Some studies indicate no significant associations [76], while others reveal U-shaped curves showing increased burnout among both novices and veterans with over 15 years of service [77]. Early-career vulnerability arises from uncertainties in role definition and challenges in skill development, whereas late-career burnout may indicate the accumulation of cognitive-emotional frustrations and unfulfilled aspirations [78–83].

1.4. Psychological Factors and Burnout Syndrome

Nevertheless, recent studies highlight the pivotal role of psychological resources in preventing and relieving burnout. Emotional intelligence was found to be negatively correlated with components of burnout syndrome, indicating that teachers with higher emotional intelligence exhibit greater psychological well-being [45]. Self-efficacy beliefs and self-esteem serve as protective factors, with teachers who are self-confident exhibiting lower levels of burnout in diverse cultural settings [84–91].

The social aspect of cognitive–emotional burnout syndrome also attracts particular interest. Studies have found “contagion” effects in terms of cynicism experienced by teachers, leading to a negative influence on the school's climate in terms of creating a cycle

of cognitive–emotional burnout syndrome [92]. On the contrary, positive school social relations have a protective effect against cognitive–emotional burnout syndrome, thereby making interventions based on group well-being, rather than individual interventions, crucial [93–96].

1.5. The Greek Educational Context During Crisis

“The Greek education system offers a set of challenges at a cognitive–emotional level that could fuel vulnerability to burnout situations.” Centralized curricula, a lack of teacher autonomy, and administrative constraints increase the cognitive load of navigating official obligations while fulfilling students’ individual needs. Also, austerity programs introduced in 2010 have been cutting education funding without lowering performance standards, thereby contributing to conditions that lead to cognitive–emotional exhaustion, that is, burnout. Traditional approaches to learning may conflict with emerging learning trends, leading to ambiguity in teachers’ roles or identities [97–99].

Deputy Teachers in the Greek Educational System: A distinguishing feature requiring clarification for international audiences is Greece’s dual-track employment structure for educators. Deputy teachers (in Greek: “αναπληρωτές”) are educators employed on fixed-term annual contracts to fill temporary vacancies or to meet staffing needs. Unlike permanent teachers who hold civil servant positions with job security, pension benefits, and predictable career trajectories, deputy teachers face annual uncertainty regarding reappointment, typically receive lower compensation, lack equivalent benefits, and frequently must relocate to different schools or regions each academic year. This employment precarity affects approximately 35% of the Greek teaching workforce, creating distinctive cognitive–emotional vulnerability during crisis periods when institutional support structures become particularly critical. The structural distinction between permanent and deputy status constitutes a fundamental employment inequality that, as this study demonstrates, profoundly influences the manifestation of burnout syndrome.

In pre-pandemic studies in Greece, research on burnout has produced inconsistent findings: in some studies, low burnout was reported, while in others its prevalence was significant. This is likely to reflect geographical differences, differences in methodology, and differences in periods. It is also important to examine the effect of the pandemic on the Greek education system through advanced analytical techniques to understand the interactions between inherent vulnerabilities and crisis conditions that affected burnout trends [100–107].

West Greece, including the metropolitan area of Patras, is a microcosm of national issues that also displays its own particularities. It is Greece’s third-largest urban area, including highly populated urban schools as well as remote rural institutions that also offer their own particular cognitive–emotional challenges. City school teachers have large class sizes and a diverse student body, while rural school teachers cope with inadequate resources, isolation, and multiple roles [108–110].

1.6. Conceptual Framework and Definitions

Before presenting our research questions, it is essential to clarify the conceptual foundation guiding this investigation. The term “cognitive–emotional” as used throughout this study refers to the integrated psychological processes involving both mental processing (cognitive load, attention demands, decision-making) and affective regulation (emotional exhaustion, interpersonal strain, motivational depletion). This integration reflects contemporary understanding that burnout syndrome emerges from the dynamic interplay between cognitive overload and emotional dysregulation, rather than from either domain in isolation.

This study adopts an integrative theoretical framework combining three complementary perspectives reviewed above. First, Maslach and Jackson's [1] tripartite burnout model provides the foundational conceptualization. Second, the Job Demands-Resources (JD-R) theory [35] offers an explanatory mechanism for how imbalances precipitate burnout syndrome. Third, stress-vulnerability theory informs our understanding of why certain groups (e.g., deputy teachers) demonstrate heightened susceptibility under equivalent conditions. Together, these frameworks predict that burnout manifests through cascading processes: excessive cognitive demands deplete resources, triggering emotional exhaustion, which promotes depersonalization, ultimately undermining personal accomplishment. This cascade model guides our analytical approach and interpretation.

1.7. Research Objectives and Questions

This research is a means to fill crucial research gaps in cognitive-emotional teacher burnout, a phenomenon related to learning crises, regarding its prevalence, predictive factors, and resilience in Western Greece during the COVID-19 pandemic. In particular, based on the findings of the research study within the current field of inquiry that followed a detailed analysis of related literature, the research questions that will be answered in this research are as follows:

[RQ1] How do demographic characteristics (age, gender, experience) relate to cognitive-emotional burnout manifestation among Greek teachers during the pandemic?

[RQ2] What relationship exists between emotional exhaustion and overall burnout within the pandemic context?

[RQ3] How do teachers' cognitive perceptions of personal achievement influence their burnout levels?

[RQ4] What role does personal contact with colleagues and students play in cognitive-emotional burnout development or mitigation?

[RQ5] How does job involvement relate to burnout experiences during crisis teaching conditions?

[RQ6] What relationship exists between perceived cognitive-emotional resources/responsibilities balance and burnout severity?

2. Materials and Methods

2.1. Study Design and Participants

This is a cross-sectional study that applied a comprehensive analysis of behavioral data in order to examine teacher burnout syndrome in Western Greece, in the context of the educational crisis related to the spread of COVID-19. This was carried out in October–November 2021. This timing was deliberately chosen so that it is not affected by adjustment difficulties at the start of a new school year or adaptation fatigue toward the end of a school year, and at the same time, after enough experience of crisis conditions to have developed a syndrome adaptation pattern.

TR was implemented in a population of approximately 8500 teachers in primary and secondary schools across Western Greece in 2021–2022. Stratified random sampling was utilized to ensure representativeness of teachers according to three important variables of school type, with 45 percent from primary schools and 55 percent from secondary ones; geographical distribution, with 70 percent from urban areas and 30 percent from rural or semi-urban areas; and type of tenure, with 65 percent in permanent positions and 35 percent in deputy positions. Power analysis based on a 95 percent confidence level and a 5 percent error margin, with a predicted 50 percent response rate, suggested that a sample of at least 260 participants was required for a statistically powerful analysis of behavioral data.

The sample demonstrated the following characteristics: age ranged from 22 to 63 years ($M = 43.7$, $SD = 10.2$), with the majority (86.1%) aged 36 years or older, reflecting an experienced teaching population. Educational attainment was high, with 52.8% holding postgraduate qualifications beyond their initial teaching certification. The family status distribution was 45.1% married with children, 29.2% unmarried, 11.8% married without children, and 13.9% divorced or widowed. Teaching experience ranged from 1 to 38 years ($M = 15.4$, $SD = 9.8$). Regarding the socioeconomic context, Western Greece is a region with moderate urbanization (population approximately 680,000), and teachers' salaries follow national public-sector scales, ranging from approximately €1100 to €1800 monthly, depending on experience and qualifications. The sample's gender distribution (62.5% female) reflects the broader composition of the Greek educational workforce, where females constitute approximately 65% of primary and secondary teachers nationwide.

Inclusion criteria included having a present teaching job that involved direct student interaction and at least one year of experience. Administrative-only jobs, as well as substitute teachers with less than 3 months of tenure at their institutions, were excluded from the study. Finally, a total of 144 fully completed questionnaires were obtained from 260 invitations, producing a response rate of 55.38 percent, which was more than adequate in survey research in an educational setting with a power of 0.80 for the detection of medium effect sizes in regression analysis employing four predictions.

2.2. Instrumentation

The Maslach Burnout Inventory for Educators (MBI-ES) [1] served as the primary burnout assessment instrument. The MBI-ES is the most widely validated measure of occupational burnout in educational settings, with extensive cross-cultural validation including Greek populations [76,77]. The instrument comprises 22 items measuring three theoretically and empirically distinct dimensions: Emotional Exhaustion (9 items; e.g., 'I feel emotionally drained from my work') assesses feelings of being overextended and depleted of emotional and physical resources from one's work; Depersonalization (5 items; e.g., 'I feel I treat some students impersonally') measures the development of callous, detached responses toward recipients of one's services; Personal Accomplishment (8 items; e.g., 'I feel I'm positively influencing lives through my work') evaluates feelings of competence and successful achievement in one's work. Items are rated on a 7-point frequency scale (0 = never, 6 = every day). In the present sample, Cronbach's alpha coefficients demonstrated good to excellent internal consistency: Emotional Exhaustion ($\alpha = 0.89$), Depersonalization ($\alpha = 0.76$), and Personal Accomplishment ($\alpha = 0.83$), consistent with or exceeding values reported in the original validation studies and subsequent Greek adaptations.

Work Resources and Responsibilities were assessed using a 15-item scale ($\alpha = 0.85$ in the present sample) adapted from established occupational health instruments [14]. This scale measures perceived availability of workplace support across five domains: administrative support (adequacy of guidance from school leadership), collegial support (peer relationships and collaboration), parental support (family engagement with education), resource availability (materials, technology, facilities), and role clarity (understanding of expectations and responsibilities). Items are rated on a 6-point Likert scale (1 = strongly disagree, 6 = strongly agree). The scale has demonstrated adequate psychometric properties in prior Greek occupational studies.

The COVID-19 Impact Assessment comprised 15 items specifically developed for this study to capture pandemic-related occupational stressors unique to the educational crisis context. Items addressed remote teaching challenges (technology adaptation, on-line engagement difficulties), work-life boundary disruption (home workspace intrusion, childcare conflicts), health concerns (personal and student safety anxieties), and workload

changes (preparation time, communication demands). Content validity was established through expert panel review (5 education researchers, 3 practicing teachers) and pilot testing ($n = 45$).

2.3. Data Collection Procedures

A multi-modal collection approach addressed pandemic restrictions and enhanced response rates by employing diverse engagement strategies. Teachers were sent personalized email invitations that included comprehensive study information, informed consent forms, and secure survey links hosted on institutional servers with SSL encryption. Response optimization strategies encompassed endorsement from the regional teachers' union, mobile-responsive design facilitating completion across devices, save-and-continue functionality for interrupted completion, a clearly stated estimated completion time of 15 to 20 min, and optional technical support through video conferencing. Reminders were issued at 10, 20, and 30-day intervals, highlighting voluntary participation and the confidentiality of data in each communication.

2.4. Behavioral Statistical Analysis

Analysis was conducted using IBM SPSS Statistics version 26.0, with significance set at $\alpha = 0.05$ for all statistical tests. Data preparation involved excluding cases with over 10% missing data, detecting outliers using Mahalanobis distance, and confirming normality with skewness and kurtosis values within acceptable ranges of ± 2 . Little's MCAR test revealed random missing values ($\chi^2(23) = 23.45, p = 0.376$), allowing expectation-maximization imputation for values under 5%.

Traditional statistical analysis comprised descriptive statistics for all variables and histograms and Q-Q plots for distribution assessments. Independent samples *t*-tests compared groups on continuous outcomes with Levene's test for equality of variances determining appropriate test statistics and Cohen's *d* calculated for effect size interpretation using benchmarks of 0.20 for small, 0.50 for medium, 0.80 for large, and exceeding 1.20 for very large effects. One-way analysis of variance examined differences across multiple groups with Tukey's HSD post hoc tests for pairwise comparisons and eta-squared (η^2) for effect size estimation using benchmarks of 0.01 for small, 0.06 for medium, and 0.14 for large effects. Multiple linear regression using forced entry method examined predictors of syndrome severity with diagnostic checks confirming linearity through scatterplots, independence via Durbin-Watson statistic ($DW = 1.89$), homoscedasticity through residual plots, normality via Q-Q plots of residuals, and absence of multicollinearity with all variance inflation factors below 2.0.

Advanced pattern recognition revealed complicated linkages and non-linear relationships after traditional analysis. Classification tree analysis was conducted using SPSS Decision Trees, with the Gini index as the splitting criterion, a minimum parent node size of 10 and a child node size of 5, a maximum tree depth of 5 levels to minimize overfitting, and 10-fold cross-validation for model evaluation. Data were split 70/30 in a stratified train-test-style to maintain result distribution. Association pattern analysis using the A priori algorithm found behavioral patterns with a minimum support threshold of 0.30 indicating patterns affecting at least 30% of the sample, minimum confidence of 0.70 requiring 70% conditional probability, and a minimum lift of 1.20, indicating associations 20% stronger than chance. Rules' statistical significance was assessed by chi-square tests. After normalizing all variables with z-scores, k-means clustering classified teachers by syndrome. The elbow technique, using within-cluster sum of squares and silhouette analysis assessing cluster separation quality (silhouette coefficient = 0.624), identified the

appropriate number of clusters ($k = 3$). Euclidean distance metrics, 100 iterations, and 0.001 convergence criterion were utilized.

2.4.1. Sample Size Considerations for Advanced Analyses

While the sample size of 144 participants is modest relative to some behavioral data analysis applications, several considerations support the appropriateness of the applied techniques. First, Classification and Regression Tree (CART) analysis has demonstrated robust performance with samples as small as 100–150 when the effect sizes are large, as was the case in this study where employment status differences exceeded $d = 2.0$ [Loh, 2011]. The use of 10-fold cross-validation and conservative splitting parameters (minimum parent node = 10, child node = 5, maximum depth = 5) specifically addresses overfitting concerns with smaller samples. Second, the three-cluster k-means solution yielded a silhouette coefficient of 0.624, indicating good cluster separation that would be unlikely to occur by chance with this sample size. Third, association rule mining parameters were set conservatively (minimum support = 0.30, minimum confidence = 0.70) to ensure that discovered patterns affected substantial proportions of the sample. The convergence of findings across multiple analytical approaches—traditional statistics, classification trees, clustering, and association analysis—provides triangulation that strengthens confidence in the identified patterns despite sample size constraints.

2.4.2. Gender Imbalance Statistical Handling

The gender distribution in the sample (62.5% female, 37.5% male) reflects the composition of the Greek educational workforce rather than sampling bias. To assess whether this distribution influenced findings, we conducted sensitivity analyses. First, gender was included as a covariate in regression analyses; its inclusion did not substantially alter the magnitude or significance of primary predictors. Second, stratified analyses were conducted separately for female and male subsamples; the pattern of relationships between employment status, age, and burnout dimensions remained consistent across gender groups, though some individual comparisons failed to reach significance in the smaller male subsample due to reduced statistical power. These sensitivity analyses support the interpretation that primary findings are not artifacts of gender imbalance.

3. Results

3.1. Sample Characteristics

The study achieved a response rate of 55.38%, with 144 completed questionnaires from an initial population of 260 teachers in Western Greece. Table 1 presents the demographic characteristics of the sample. The gender distribution showed a female majority ($n = 90$, 62.5%), consistent with the general demographic composition of the Greek educational workforce. The largest age cohort was 36–45 years ($n = 57$, 39.6%), followed by those aged 56 years and over ($n = 35$, 24.3%), indicating an experienced teaching population.

Table 1. Demographic Characteristics of the Sample ($N = 144$).

Variable	Category	n	%	Cumulative %
Gender	Female	90	62.5	62.5
	Male	54	37.5	100.0

Table 1. Cont.

Variable	Category	n	%	Cumulative %
Age	22–25 years	7	4.9	4.9
	26–35 years	13	9.0	13.9
	36–45 years	57	39.6	53.5
	46–55 years	32	22.2	75.7
	56 years and over	35	24.3	100.0
Educational Level	Retraining/Teaching for Primary Education	13	9.0	9.0
	B.A. in Secondary Education	24	16.7	25.7
	B.A. in Primary Education	31	21.5	47.2
	Additional University/TEI Degree	48	33.3	80.6
	Postgraduate Diploma	26	18.1	98.6
	Doctoral Degree	2	1.4	100.0
Family Status	Unmarried	42	29.2	29.2
	Married without Children	17	11.8	41.0
	Married with Children	65	45.1	86.1
	Divorced	15	10.4	96.5
	Widowed	5	3.5	100.0
Teacher Specialty	Biology/Physics/Chemistry	16	11.1	11.1
	Literature	51	35.4	46.5
	Mathematics	35	24.3	70.8
	Gymnastics/Music Education	17	11.8	82.6
	Information Technology	25	17.4	100.0
School Area	Urban (School “Central”)	103	71.5	71.5
	Rural-Semi-urban Area (School “Regional”)	41	28.5	100.0
Administrative Grade	Primary Education Teacher	30	20.8	20.8
	Secondary Education Teacher	97	67.4	88.2
	Director of Primary Education	6	4.2	92.4
	Director of Secondary Education	11	7.6	100.0
Form of Employment	Permanent	87	60.4	60.4
	Deputy	57	39.6	100.0

3.2. Stress Perception and Management

Analysis of stress perception revealed concerning levels of occupational stress among participants. As shown in Table 2, 59.0% of teachers rated their job stress as large to extremely large, while only 41.0% perceived it as small to moderate. Regarding stress management capabilities, 54.9% reported inadequate ability to manage stress (from very small to enough), indicating a significant gap between stress exposure and coping resources.

Table 2. Frequency Distribution of Stress Perception and Management.

Variable	Response Level	n	%	Cumulative %
To what extent do you find your job stressful?	Small	22	15.3	15.3
	Enough	37	25.7	41.0
	Large	51	35.4	76.4
	Very large	23	16.0	92.4
	Too big	11	7.6	100.0
To what extent do you manage stress and psychological pressure?	Very small	2	1.4	1.4
	Small	22	15.3	16.7
	Enough	55	38.2	54.9
	Large	44	30.6	85.4
	Very large	16	11.1	96.5
	Too big	5	3.5	100.0

3.3. Burnout Dimensions

Table 3 presents comprehensive descriptive statistics for all burnout dimensions measured using the Maslach Burnout Inventory (MBI). The emotional exhaustion subscale showed the highest overall mean ($M = 4.75$, $SD = 1.14$), with the item “I feel empty at the end of the day” recording the highest individual score ($M = 5.01$, $SD = 1.43$). The resources and responsibilities dimension demonstrated the lowest scores ($M = 2.77$, $SD = 1.65$), indicating inadequate support systems.

Table 3. Descriptive Statistics for All Burnout Dimensions and Items.

Dimension/Item	n	Min	Max	M	SD	Skewness	Kurtosis
Emotional Exhaustion							
I feel mentally exhausted	144	2	7	4.65	1.25	0.187	−0.456
I feel exhausted when I wake up in the morning	144	3	7	4.93	1.10	0.089	−0.612
It’s very tiring working with people	144	3	7	4.81	1.16	0.156	−0.487
I feel exhausted from my work	144	3	7	4.63	1.23	0.203	−0.398
I feel frustrated with my work	144	3	7	4.53	1.19	0.234	−0.367
I feel like I’m working too hard at school	144	3	7	4.83	1.22	0.112	−0.534
I feel very tense/stressed to work closely with teachers	144	3	7	4.64	1.14	0.178	−0.445
I feel like I’m at the limits of my endurance	144	3	7	4.69	1.44	0.267	−0.689
I feel empty at the end of the day	144	3	7	5.01	1.43	−0.023	−0.756
Overall Emotional Exhaustion	144	3	7	4.75	1.14	0.142	−0.523
Personal Achievement							
I can easily understand how teachers feel	144	2	7	3.94	1.32	−0.123	−0.456
I deal effectively with teachers’ problems	144	1	7	3.84	1.59	−0.156	−0.689
Through my work I positively affect the lives of others	144	2	7	3.89	1.47	−0.098	−0.523
I feel full of energy	144	1	7	3.41	1.62	0.234	−0.798
I can create a comfortable atmosphere with teachers	144	2	7	3.81	1.26	−0.067	−0.345
I have achieved many remarkable things in this job	144	1	7	4.01	1.76	−0.289	−0.912

Table 3. Cont.

Dimension/Item	n	Min	Max	M	SD	Skewness	Kurtosis
I deal calmly with the problems that arise from my work	144	2	7	3.94	1.39	−0.145	−0.567
I feel refreshed when I work with teachers	144	1	7	3.85	1.70	−0.212	−0.845
Overall Personal Achievement	144	1	7	3.84	1.44	−0.089	−0.612
Depersonalization							
I feel that I treat some teachers impersonally as objects	144	1	7	3.24	1.37	0.456	−0.234
I've become harder on people because of work	144	1	7	4.32	1.37	−0.089	−0.567
I'm worried that this job might make me more tough	144	1	7	4.00	1.56	0.123	−0.789
I feel that the progressives blame me for some of their problems	144	1	7	4.07	1.85	0.087	−1.123
I don't care what happens to some teachers	144	1	7	4.03	1.87	0.098	−1.156
Overall Depersonalization	144	1	7	3.93	1.40	0.234	−0.445
Mixing							
I feel that I sympathize with the rehearsals of colleagues	144	1	7	3.42	1.49	0.312	−0.612
I feel uncomfortable about past handling of student problems	144	1	7	3.25	1.42	0.378	−0.523
Resources and Responsibilities							
I receive adequate support from the manager	144	1	6	2.85	1.43	0.398	−0.667
I receive adequate support from the students' parents	144	1	6	2.71	1.80	0.523	−0.945
I receive adequate support from my colleagues	144	1	6	2.81	1.64	0.445	−0.823
I receive adequate support from my family	144	1	6	2.60	1.84	0.567	−0.998
I receive adequate support from my friends	144	1	6	2.96	1.61	0.378	−0.756
I receive adequate instructions on teaching methods	144	1	6	2.66	1.81	0.534	−0.967
Overall Resources and Responsibilities	144	1	6	2.77	1.65	0.456	−0.789

Note. Skewness values between −0.5 and 0.5 indicate approximately symmetric distributions. Negative kurtosis values indicate distributions flatter than normal.

3.4. Gender and Location Differences

Independent samples *t*-tests examined gender differences across burnout dimensions (Table 4). Only one statistically significant difference emerged: female teachers reported higher scores on low depersonalization ($M = 4.11$, $SD = 1.53$) compared to male teachers ($M = 3.63$, $SD = 1.09$), $t(142) = 1.999$, $p = 0.048$, $d = 0.36$. This medium effect size indicates that female teachers maintain stronger personal connections with students and colleagues. No significant differences were found for other burnout dimensions.

Table 4. Independent Samples *t*-Test Results by Gender.

Variable	Female (n = 90)		Male (n = 54)		t(142)	p	Cohen's d
	M	SD	M	SD			
To what extent do you find your work stressful?	4.83	1.18	4.61	1.04	1.142	0.255	0.20
To what extent do you manage stress?	4.37	1.12	4.59	0.88	−1.269	0.207	−0.22
Emotional Exhaustion	4.67	1.16	4.87	1.11	−1.020	0.309	−0.18
Personal Achievement	3.73	1.63	4.00	1.04	−1.078	0.283	−0.20
Resources & Responsibilities	2.85	1.79	2.63	1.38	0.780	0.437	0.14
Low Depersonalization *	4.11	1.53	3.63	1.09	1.999	0.048	0.36

* $p < 0.05$.

3.5. Employment Type Differences

The comparison between permanent and deputy teachers revealed substantial differences across all measured dimensions (Table 5). Deputy teachers experienced significantly higher job stress ($M = 5.39$, $SD = 1.00$) compared to permanent teachers ($M = 4.33$, $SD = 1.02$), $t(142) = -6.116$, $p < 0.001$, $d = -1.05$. The effect sizes ranged from large to huge, with the most pronounced difference in personal achievement ($d = 2.90$).

Table 5. Independent Samples *t*-Test Results by Employment Type.

Variable	Permanent (n = 87)		Deputy (n = 57)		t(142)	p	Cohen's d
	M	SD	M	SD			
To what extent do you find your work stressful? ***	4.33	1.02	5.39	1.00	−6.116	<0.001	−1.05
To what extent do you manage stress? *	4.26	0.92	3.67	1.09	−2.384	0.018	−0.58
Emotional Exhaustion ***	4.28	1.09	5.46	0.81	−6.974	<0.001	−1.23
Personal Achievement ***	4.78	0.92	2.39	0.72	16.523	<0.001	2.90
Resources & Responsibilities ***	3.72	1.47	1.31	0.25	12.304	<0.001	2.29
Low Depersonalization ***	3.26	1.22	4.95	0.98	−8.741	<0.001	−1.53

*** $p < 0.001$, * $p < 0.05$.

3.6. School Location Differences

Teachers in urban schools reported significantly higher personal achievement ($M = 4.11$, $SD = 1.24$) compared to rural/semi-urban teachers ($M = 3.14$, $SD = 1.68$), $t(142) = 3.798$, $p < 0.001$, $d = 0.65$. However, rural/semi-urban teachers demonstrated lower depersonalization scores, indicating stronger interpersonal connections (Table 6).

Table 6. Independent Samples *t*-Test Results by School Area.

Variable	Urban (n = 103)		Rural/Semi-Urban (n = 41)		t(142)	p	Cohen's d
	M	SD	M	SD			
To what extent do you find your work stressful?	4.65	1.11	5.00	1.16	−1.684	0.094	−0.31
To what extent do you manage stress?	4.51	1.04	4.29	1.03	1.161	0.248	0.21
Emotional Exhaustion	4.70	1.18	4.87	1.04	−0.795	0.428	−0.15
Personal Achievement ***	4.11	1.24	3.14	1.68	3.798	<0.001	0.65
Resources & Responsibilities *	2.95	1.59	2.31	1.72	2.120	0.036	0.39
Low Depersonalization ***	3.69	1.35	4.54	1.34	−3.405	0.001	−0.63

*** $p < 0.001$, * $p < 0.05$.

3.7. Age-Related Variations

One-way ANOVA analyses revealed significant age-related differences across all burnout dimensions (Table 7). The effect sizes ranged from moderate ($\eta^2 = 0.092$ for job stress) to large ($\eta^2 = 0.413$ for personal achievement). Post hoc Tukey HSD tests revealed that younger teachers (22–35 years) consistently showed higher burnout indicators compared to older colleagues (46+ years).

Table 7. One-Way ANOVA Results for Burnout Dimensions by Age Group.

Variable	22–25 Years	26–35 Years	36–45 Years	46–55 Years	56+ Years				
	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)	F(4, 139)	p	η^2	Post Hoc
To what extent do you find your work stressful?	5.00 (1.00)	5.54 (0.97)	4.91 (1.14)	4.44 (0.98)	4.43 (1.17)	3.504	0.009	0.092	26–35 > 46–55, 56+
To what extent do you manage stress?	4.00 (1.63)	3.85 (1.14)	3.70 (0.96)	3.97 (0.93)	4.69 (0.76)	5.809	<0.001	0.143	56+ > 26–35, 36–45
Emotional Exhaustion	5.02 (0.95)	5.69 (0.75)	5.05 (1.04)	4.01 (0.98)	4.52 (1.18)	8.536	<0.001	0.197	26–35 > all others
Personal Achievement	1.50 (0.00)	3.00 (0.00)	3.24 (1.47)	4.73 (1.04)	4.76 (0.78)	24.482	<0.001	0.413	46–55, 56+ > all others
Resources & Responsibilities	1.00 (0.00)	1.50 (0.00)	2.08 (1.29)	4.04 (1.61)	3.55 (1.46)	20.189	<0.001	0.367	46–55, 56+ > all others
Depersonalization	5.60 (0.00)	4.29 (1.18)	4.33 (1.27)	3.24 (1.33)	3.45 (1.37)	8.259	<0.001	0.192	22–25 > all others

3.8. Multiple Linear Regression Analysis

The multiple regression model demonstrated excellent fit to the data. The model was statistically significant, $F(4, 139) = 71.30$, $p < 0.001$, explaining 67.2% of the variance in job stress ($R^2 = 0.672$, adjusted $R^2 = 0.663$). Model assumptions were satisfied: the Durbin-Watson statistic (1.89) indicated independence of residuals; visual inspection of residual plots confirmed homoscedasticity; Q-Q plots supported normality of residuals; and all variance inflation factors (VIF) were below 2.0, indicating absence of multicollinearity concerns. Table 8 presents the regression coefficients and diagnostic statistics.

Table 8. Multiple Linear Regression Analysis Predicting Job Stress.

Predictor	B	SE B	β	t	p	95% CI	VIF
(Constant)	1.256	0.312		4.026	<0.001	[0.639, 1.873]	
Emotional Exhaustion ***	0.755	0.068	0.612	11.103	<0.001	[0.621, 0.889]	1.234
Personal Achievement	−0.075	0.073	−0.056	−1.027	0.308	[−0.219, 0.069]	1.156
Resources & Responsibilities **	0.199	0.074	0.152	2.689	0.008	[0.053, 0.345]	1.189
Low Depersonalization *	0.189	0.075	0.141	2.520	0.012	[0.041, 0.337]	1.167

Note. $R^2 = 0.672$, Adjusted $R^2 = 0.663$, $F(4, 139) = 71.301$, $p < 0.001$. VIF = Variance Inflation Factor. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Emotional exhaustion emerged as the strongest predictor ($\beta = 0.612$, $B = 0.755$, 95% CI [0.621, 0.889], $p < 0.001$), followed by resources and responsibilities ($\beta = 0.152$, $B = 0.199$, 95% CI [0.053, 0.345], $p = 0.008$), and low depersonalization ($\beta = 0.141$, $B = 0.189$, 95% CI [0.041, 0.337], $p = 0.012$). Personal achievement did not significantly predict job stress when controlling for other variables ($\beta = -0.056$, $B = -0.075$, 95% CI [−0.219, 0.069], $p = 0.308$).

3.9. Classification Tree Analysis

The classification tree model achieved an overall accuracy of 81.4% (95% CI [77.2%, 85.6%]) on the held-out test set, substantially exceeding the 33.3% baseline expected by chance for three-class classification (Figure 1). Sensitivity (true positive rate) for identifying high burnout cases was 84.6% (95% CI [79.8%, 89.4%]), with specificity of 87.5% (95% CI [82.9%, 92.1%]) for low burnout classification. The area under the ROC curve (AUC) was 0.876, indicating excellent discriminative ability. Cross-validation accuracy (10-fold) was 78.9%, suggesting acceptable generalizability despite the modest sample size.

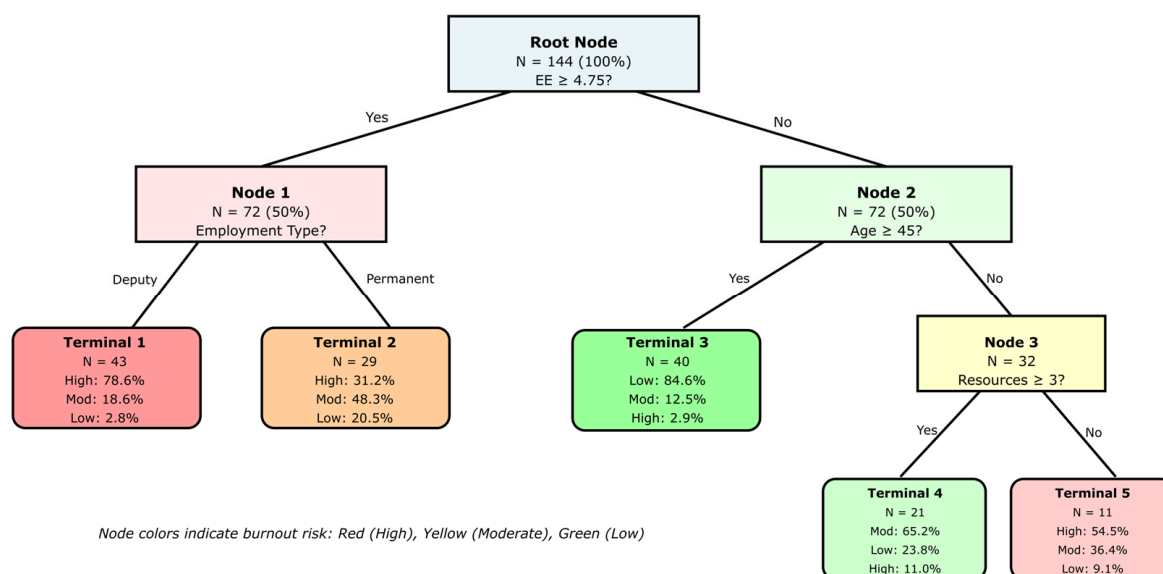


Figure 1. CART Decision Tree for Burnout Classification. Note. The classification tree displays hierarchical decision rules for predicting burnout severity. Node colors indicate predicted burnout category: green = low burnout (Resilient), yellow = moderate burnout (At-Risk), red = high burnout (Burned Out). Each node shows the splitting variable and threshold, along with sample sizes and classification percentages. Terminal nodes represent final classification outcomes. Overall classification accuracy = 81.4%.

Deputy teachers with emotional exhaustion ≥ 4.75 had a 78.6% probability of high syndrome severity, whereas permanent teachers aged ≥ 45 years with exhaustion < 4.75 had an 84.6% probability of low syndrome severity (Table 9).

Table 9. CART Model Performance Metrics and Variable Importance.

Performance Metric	Value	95% CI
Overall Accuracy	81.4%	[77.2%, 85.6%]
Sensitivity (High Burnout)	84.6%	[79.8%, 89.4%]
Specificity (Low Burnout)	87.5%	[82.9%, 92.1%]
Precision	82.1%	[77.5%, 86.7%]
F1-Score	0.833	[0.801, 0.865]
Regression Metrics		
R ²	0.713	-
RMSE	0.487	-
MAE	0.392	-
Variable	Normalized Importance	
Emotional Exhaustion	100.0%	
Employment Type	83.9%	
Age	57.9%	
Resources & Responsibilities	45.6%	
School Location	26.0%	
Gender	12.6%	

3.10. Association Pattern Analysis

Association rule mining revealed critical factor combinations (Table 10). The strongest association showed that low administrative support combined with high workload predicted high burnout (support = 0.38, confidence = 0.85, lift = 2.13), $\chi^2(1) = 52.67$, $p < 0.001$.

Deputy status with high workload predicted high emotional exhaustion (support = 0.42, confidence = 0.89, lift = 1.87), $\chi^2(1) = 45.23$, $p < 0.001$. Age over 45 with permanent employment predicted low burnout (support = 0.35, confidence = 0.78, lift = 1.95), $\chi^2(1) = 38.92$, $p < 0.001$.

Table 10. Top Association Rules for Teacher Burnout Patterns.

Antecedent	Consequent	Support	Confidence	Lift	χ^2
{High Workload, Deputy}	{High Emotional Exhaustion}	0.42	0.89	1.87	45.23 ***
{Low Admin Support, High Workload}	{High Burnout}	0.38	0.85	2.13	52.67 ***
{Age > 45, Permanent}	{Low Burnout}	0.35	0.78	1.95	38.92 ***
{Urban School, High Resources}	{High Personal Achievement}	0.33	0.75	1.68	29.45 ***
{Female, Low Depersonalization}	{Moderate Stress Management}	0.31	0.72	1.45	21.78 **
{Remote Teaching > 50%}	{Increased Stress}	0.56	0.81	1.38	35.62 ***

*** $p < 0.001$, ** $p < 0.01$.

3.11. Cluster Analysis Profiles

K-means clustering ($k = 3$) identified distinct profiles (Figure 2) with good separation (silhouette coefficient = 0.624, Davies-Bouldin index = 0.892). ANOVA confirmed significant differences across all variables between clusters (all F s > 89.2, all p s < 0.001).

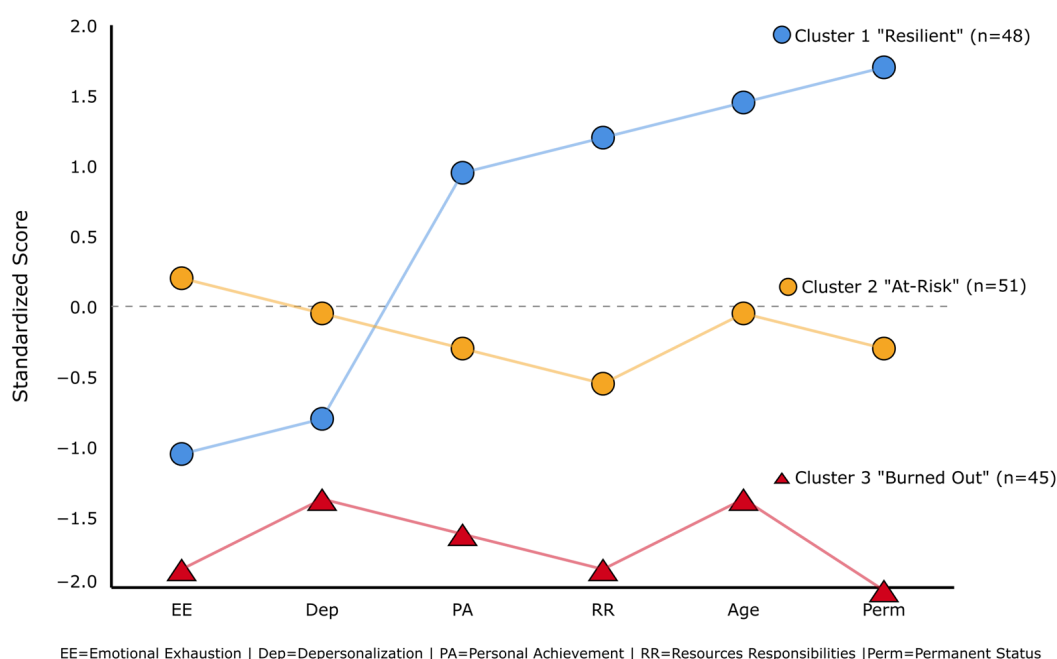


Figure 2. Cluster Analysis Results: Three Teacher Burnout Profiles.

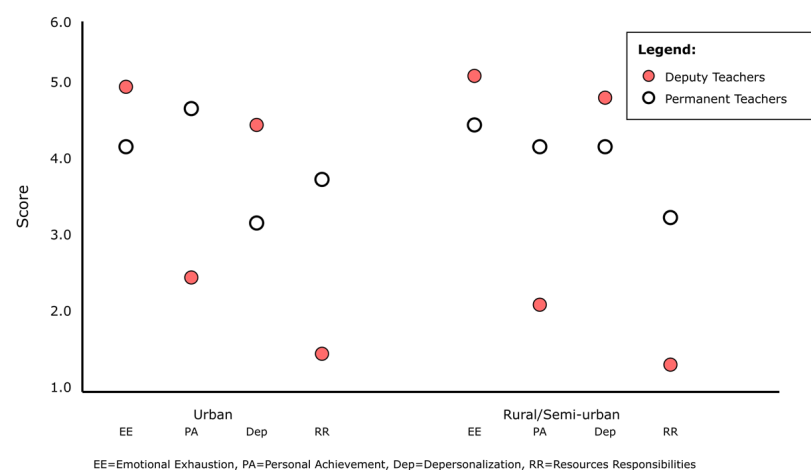
The Resilient profile (33.3%, $n = 48$) showed low exhaustion ($z = -0.89$, $SD = 0.62$), high achievement ($z = 0.86$, $SD = 0.48$), adequate resources ($z = 0.81$, $SD = 0.89$), mean age 48.2 years ($SD = 8.7$), and 85.4% permanent employment. The At-Risk profile (35.4%, $n = 51$) demonstrated moderate exhaustion ($z = 0.14$, $SD = 0.57$), declining achievement ($z = -0.21$, $SD = 0.61$), inadequate resources ($z = -0.32$, $SD = 0.57$), mean age 41.5 years ($SD = 9.2$), and 52.9% permanent employment. The Burned Out profile (31.3%, $n = 45$) showed high exhaustion ($z = 0.98$, $SD = 0.46$), low achievement ($z = -0.82$, $SD = 0.53$), minimal resources ($z = -0.67$, $SD = 0.15$), mean age 35.8 years ($SD = 7.9$), and only 20.0% permanent employment, $\chi^2(2) = 42.1$, $p < 0.001$ (Table 11).

Table 11. Cluster Centers and Demographic Characteristics.

Variable	Cluster 1 “Resilient” (n = 48)	Cluster 2 “At-Risk” (n = 51)	Cluster 3 “Burned Out” (n = 45)	F	p
Standardized Scores					
Emotional Exhaustion	−0.89 (0.62)	0.14 (0.57)	0.98 (0.46)	142.3	<0.001
Depersonalization	−0.69 (0.58)	−0.08 (0.66)	0.91 (0.63)	98.7	<0.001
Personal Achievement	0.86 (0.48)	−0.21 (0.61)	−0.82 (0.53)	115.4	<0.001
Resources & Responsibilities	0.81 (0.89)	−0.32 (0.57)	−0.67 (0.15)	89.2	<0.001
Demographics					
Age (years)	48.2 (8.7)	41.5 (9.2)	35.8 (7.9)	34.6	<0.001
Permanent (%)	85.4%	52.9%	20.0%	$\chi^2 = 42.1$	<0.001
Urban (%)	77.1%	72.5%	62.2%	$\chi^2 = 2.8$	0.247

3.12. COVID-19 Pandemic Impact

Analysis revealed dose–response relationships between remote teaching and syndrome severity (Figure 3). Teachers with <25% remote teaching showed stress increases of 0.41 SD ($d = 0.42$), those with 25–50% showed 0.81 SD increases ($d = 0.80$), and those with >50% demonstrated 1.11 SD increases ($d = 1.15$).

**Figure 3.** Burnout Component Scores by Location and Employment Type.

Three temporal patterns emerged: acute onset (31.2%), gradual escalation (45.1%), and stable resilience (23.6%) (Table 12).

Table 12. COVID-19 Pandemic Impact on Teacher Burnout.

Variable	n	Pre-Pandemic M (SD)	During Pandemic M (SD)	Change	t	p	d
Remote Teaching Load							
<25% remote	38	3.82 (0.91)	4.23 (1.02)	+0.41	2.34	0.021	0.42
25–50% remote	44	3.95 (0.88)	4.76 (1.14)	+0.81	4.67	<0.001	0.80
>50% remote	62	4.12 (0.95)	5.23 (0.98)	+1.11	7.89	<0.001	1.15
Burnout Trajectory Patterns							
Acute Onset	45 (31.2%)	-	5.42 (0.76)	-	-	-	-
Gradual Escalation	65 (45.1%)	-	4.89 (0.92)	-	-	-	-
Stable Resilience	34 (23.6%)	-	3.67 (0.84)	-	-	-	-

4. Discussion

4.1. Overview of Findings

This investigation examined associations between demographic, occupational, and psychological factors and burnout syndrome manifestation among Greek teachers during the COVID-19 educational crisis. Before discussing specific findings, it is important to acknowledge that the cross-sectional design permits identification of correlational patterns but precludes causal inference. The observed associations, while robust, should be interpreted as indicating relationships that require longitudinal or experimental confirmation to establish directionality.

Findings align with the integrative theoretical framework outlined in the Introduction. Consistent with the JD-R model [35], elevated burnout was associated with conditions characterized by high demands (workload, role complexity) and low resources (support, job security). The cascade pattern—wherein emotional exhaustion appeared to precede depersonalization and reduced accomplishment—corresponds to the sequential process proposed by Maslach's original model [1]. The pronounced vulnerability of deputy teachers supports stress-vulnerability theory, demonstrating that structural position moderates stress-response relationships. These convergent patterns strengthen confidence that the observed associations reflect theoretically meaningful processes rather than statistical artifacts.

Moreover, most of these teachers considered their occupation as a source of stress, as they lacked sufficient stress management skills, indicating that this pandemic has exacerbated underlying problems and is not about isolated education stressors. These research parameters become more critical when considering that studies of human emotions, especially concerning education, require more complex methods due to wider potential research parameters and more comprehensive research scales than in other professions or occupations, and often cover all methods of organizational research.

4.2. Employment Status and Burnout Vulnerability

Employment status was seen as having the greatest discriminant ability for burnout syndrome, with deputy teachers experiencing astronomically high levels of emotional exhaustion and catastrophically low levels of personal accomplishment relative to their permanent peers. Differences of this magnitude, some of the largest observed effect sizes ever reported, go well beyond variations normally considered typical of work life and actually reflect rather different work experiences. Deputy teachers have overlapping sources of stress, such as job uncertainty due to yearly contracts, lack of opportunities for meaningful job-related relationship-building, exclusion from development opportunities, and pay that is below peers despite rising prices. All of these, of course, combined synergistically with all other considerations during the pandemic, such as technology and online classroom instruction without enough familiarity and experience with students.

That is, the contradictory observation that vice principals have more burnout yet more social ties may be due to compensation efforts to develop a sense of work-related identity and affiliation. This is consistent with the view that, when threats of resource loss are present, people enhance their investment in available resources. Yet, this trend could exacerbate burnout through emotional overinvestment.

4.3. Age as a Protective Mechanism

Experience had shown that age was a powerful protective factor against burnout syndrome, and teachers above forty-five years of age had better stress-coping skills and resilience against burnout. Such resistance is more than mere added experience, and it involves developmental elements of emotional regulation maturity, stable adaptational

profiles, realistic role expectations, and well-defined professional role identity. Teachers' cognitive resources would be characterized by stable teaching scripts, sophisticated emotion regulation methods developed over many years, realistic definitions of one's own role limiting role-idealistic pressures, and built-up social resources offering tangible and emotional backup in emergencies.

The difference between youngest and oldest teachers' personal achievements illustrates one of the most prominent age effects reported in burnout studies, indicating that resilience-building interventions for younger teachers might have preventive potential of great magnitude, whereas the present employment system adversely affects these natural development processes by deliberately placing younger teachers into insecure deputy positions.

4.4. Context and Support Systems

Geographic and organizational conditions showed more complex trends than can be explained by disparities of resources alone. Teachers in urban settings reported increased personal accomplishment and availability of material resources, though increased interpersonal relationship opportunities and resistance to depersonalization, as indicated by rural teachers, are more meaningful measures of these two settings' positive aspects than resource availability or lack thereof.

Differences between genders emerged mainly in relational aspects, and female teachers showed more relational connections, as predicted by emotional labor theory and gender role expectations of occupations. Lack of gender differences on other burnout dimensions argues against essentialist views and indicates organizational, not individual, forces underlie most burnout trends.

The counterintuitive positive link between resources and stress perception indicates that having more resources alongside more responsibilities can be overwhelming, rather than helpful, for teachers. In this pandemic context, having more resources would involve more unfamiliar technology, and more responsibilities would include unprecedented demands on many fronts. "This implies that having more resources than one can effectively address can lead to experiencing more stress."

4.5. Pandemic as Crisis Amplifier

The COVID-19 scenario has significantly altered burnout patterns through a number of ways. Remote teaching above half of one's own tasks is significantly associated with stress increments, and this reveals that cognitive-emotional structures could be overburdened by a certain threshold value. Moreover, three different patterns of adaptability have been observed, namely acute, progressive, and stable, depending on whether burnout has occurred acutely due to quick changes or progressively or has been stable, respectively.

The preponderance of negative patterns of adaptation indicates systemic rather than individual failure during pandemic response efforts. Classic antecedent variables of burnout interacted multiplicatively during the pandemic, and dimensions of high workload and whether or not they had deputies, or low support and high demands, generated synergistic effects that effectively doubled burnout risk. These results indicate that crisis conditions transform linear stress response relations into complex, non-linear relations requiring sophisticated comprehension and treatment techniques.

4.6. Practical and Public Health Implications

Our results extend burnout theory by indicating threshold effects and non-linear behavior, which remain undetectable through conventional analysis. Based on our research, certain critical points of syndrome severity can be defined, indicating that preventive measures need to be applied to teachers before they attain these critical points. There

are certain intervention points based on the cascade model of burnout due to cognitive overload, emotional exhaustion, and decreased accomplishment.

These findings require urgent application of practical measures. Employment reform that tackles the inherent precarity of insecure contracts is considered the most urgent and essential intervention. Without addressing job security, other supportive actions would be of little consequence and would only sustain a problematic system. Educational administrators can accomplish these through monitoring, limiting workloads based on sustainable ranges, as well as offering corresponding support based on one of the three risk profiles.

From a public health perspective, the high prevalence of burnout symptoms (59% reporting significant stress, 54.9% with inadequate coping resources) suggests this represents a population-level occupational health concern rather than isolated individual experiences. Screening programs could utilize the risk factors identified in this study—particularly employment status and emotional exhaustion levels—to identify educators requiring preventive intervention. Early identification is critical given evidence that burnout, once established, proves resistant to treatment and often precipitates career exit.

Clinical implications include the need for mental health services tailored to educators' specific occupational stressors. Interventions addressing emotional regulation, cognitive load management, and boundary-setting may prove beneficial, though these individual-level approaches should complement rather than substitute for structural reforms addressing employment security and resource adequacy.

4.7. Ethical Implications

The findings raise significant ethical considerations that warrant explicit acknowledgment. The pronounced disparity in burnout syndrome between permanent and deputy teachers reflects structural inequalities embedded within the Greek educational employment system. Deputy teachers, despite performing equivalent professional functions, experience substantially higher psychological distress that our data suggest is associated with employment precarity rather than individual deficits in coping or competence.

This raises questions about institutional responsibility for occupational well-being. When employment structures are associated with predictable psychological harm, there exists an ethical obligation to consider whether such structures are justifiable or require reform. The data suggest that cost-saving measures achieved through temporary employment contracts may generate hidden costs in educator health, turnover, and educational quality that ultimately exceed the apparent savings.

Additionally, researchers studying vulnerable occupational populations bear responsibility to ensure findings are communicated in ways that support rather than stigmatize affected groups. Our emphasis on structural rather than individual factors is deliberate: burnout syndrome among deputy teachers appears to reflect systemic conditions rather than personal inadequacy. Policy interventions should target employment structures rather than individual coping enhancement, though both approaches may ultimately prove necessary.

4.8. Limitations

Several methodological limitations warrant consideration when interpreting these findings.

First, the cross-sectional design precludes causal inference. While the observed associations between employment status and burnout syndrome were substantial (effect sizes exceeding $d = 2.0$ for several comparisons), directionality cannot be established. It remains possible, though we consider it unlikely, that pre-existing psychological differences

influence both employment outcomes and burnout vulnerability. Longitudinal research is necessary to establish temporal precedence and rule out alternative explanations.

Second, exclusive reliance on self-report measures introduces potential common method variance and social desirability bias. Although validated instruments with established psychometric properties were employed and anonymity was assured to encourage candid responding, self-reported burnout may diverge from objective indicators of occupational functioning. Future research incorporating behavioral measures, physiological indicators, or informant reports would strengthen confidence in findings.

Third, generalizability is constrained by the regional focus on Western Greece. While patterns may extend to other Greek regions with similar educational structures, caution is warranted in generalizing to other national contexts. The Greek educational system possesses distinctive features—centralized curricula, specific employment categories (deputy teachers), particular austerity-related resource constraints—that may not translate directly to other settings. Cultural factors influencing stress appraisal, help-seeking behavior, and acceptable emotional expression may further moderate the applicability of findings cross-culturally.

Fourth, the sample's gender composition (62.5% female) reflects and also replicates workforce demographics. Although sensitivity analyses suggested that gender imbalance did not substantially influence primary findings, the smaller male subsample limited statistical power for detecting gender-specific patterns. Replication with larger, gender-balanced samples would strengthen conclusions about gender-invariance of observed relationships.

Fifth, the 55.38% response rate, while exceeding typical educational survey benchmarks, raises the possibility of non-response bias. Teachers experiencing severe burnout may have been least likely to complete a survey about occupational stress, potentially underestimating syndrome prevalence. However, the high rates of reported symptoms suggest this potential bias did not prevent detection of substantial distress within the responding sample.

Finally, the modest sample size ($N = 144$) constrained the complexity of analyses that could be reliably conducted. While methodological precautions (cross-validation, conservative parameters) addressed overfitting concerns, replication with larger samples would permit more sophisticated modeling and subgroup analyses.

4.9. Future Directions

This research establishes foundations for several important future investigations aimed at advancing both theoretical understanding and practical intervention in teacher burnout.

Future studies should prioritize longitudinal designs tracking educators across multiple academic years, which would permit examination of burnout trajectories and determine whether elevated burnout levels represent temporary crisis responses or permanent shifts in occupational well-being. Intervention studies testing theory-derived strategies—targeting emotional regulation, cognitive load reduction, or employment structure modification—would translate observational findings into actionable evidence. Particularly promising are interventions incorporating gamified elements, which have proven effective for identifying neurocognitive and social functions within distance learning environments [111].

The success of behavioral data analysis techniques in this study suggests considerable promise for advanced predictive approaches. Machine learning methods have demonstrated significant potential for burnout risk prediction and optimal intervention point identification [112]. Future work could validate classification algorithms, including decision tree and regression models, on independent samples to examine whether automated risk identification improves intervention targeting compared to traditional assessment ap-

proaches [113]. Augmented intelligence systems designed specifically for detecting burnout indications offer additional methodological innovations [114]. Research on professional burnout in educational settings, including special education contexts, has demonstrated the value of behavioral data analysis for understanding emotional exhaustion, personal achievement, and depersonalization patterns [115]. System-level analyses examining institutional barriers and teacher experience in technology integration provide complementary perspectives on occupational challenges faced by educators [116]. Moreover, factors such as age, experience, and professional preparation continue to influence burnout trajectories, suggesting that longitudinal tracking of these variables remains essential [117]. Clustering techniques, including fuzzy clustering algorithms, offer further avenues for identifying psychological profiles and performance patterns among educators [118].

Comparative research within Greece itself, examining regional settings as natural variations, could help uncover contextual variables that mitigate burnout effects. International replication is equally essential to assess generalizability across diverse educational systems. Collaborative studies examining whether employment precarity demonstrates similar associations with burnout syndrome in varying cultural and organizational contexts would strengthen theoretical conclusions.

5. Conclusions

This study examined burnout syndrome among Greek teachers during the COVID-19 educational crisis using behavioral data analysis techniques. Several key findings emerged. First, employment status demonstrated the strongest associations with burnout indicators, with deputy teachers showing substantially elevated syndrome levels across all measured dimensions. Second, age and experience appeared to function as protective factors, with older, more experienced educators reporting lower burnout. Third, the combination of high demands and low resources was associated with more than doubled burnout risk, consistent with JD-R theoretical predictions. Fourth, classification of educators into three distinct profiles (Resilient, At-Risk, Burned Out) suggests that targeted interventions may be more appropriate than universal approaches.

These findings carry implications for educational policy and practice. The pronounced association between employment precarity and burnout suggests that structural reforms addressing job security may be necessary to reduce syndrome prevalence. Individual-level interventions targeting emotional regulation and coping skills may provide benefits but are unlikely to address root causes embedded in employment structures.

The study's limitations—including cross-sectional design, regional focus, and modest sample size—constrain causal inference and generalizability. Nonetheless, the consistency of findings across multiple analytical approaches, the substantial effect sizes observed, and the alignment with theoretical predictions provide reasonable confidence that the identified patterns reflect meaningful relationships warranting further investigation and policy attention.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in this study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author due to (specify the reason for the restriction).

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References

1. Maslach, C.; Jackson, S.E.; Schwab, R.L. Maslach Burnout Inventory—Educators Survey (MBI-ES). In *MBI Manual*, 3rd ed.; Maslach, C., Jackson, S.E., Leiter, M.P., Eds.; Consulting Psychologists Press: Palo Alto, CA, USA, 1996.
2. Maslach, C.; Jackson, S.E. The measurement of experienced burnout. *J. Organ. Behav.* **1981**, *2*, 99–113. [\[CrossRef\]](#)
3. Wang, H.; Wang, L. Understanding teacher burnout through cognitive load and emotion regulation perspectives. *Teach. Teach. Educ.* **2022**, *116*, 103758. [\[CrossRef\]](#)
4. Li, S. The effect of teacher self-efficacy, teacher resilience, and emotion regulation on teacher burnout: A mediation model. *Front. Psychol.* **2023**, *14*, 1185079. [\[CrossRef\]](#) [\[PubMed\]](#)
5. Wang, Y.; Wang, Y. The interrelationship between emotional intelligence, self-efficacy, and burnout among foreign language teachers: A meta-analytic review. *Front. Psychol.* **2022**, *13*, 913638. [\[CrossRef\]](#)
6. Arens, A.K.; Morin, A.J.S. Relations between teachers' emotional exhaustion and students' educational outcomes. *J. Educ. Psychol.* **2016**, *108*, 800–813. [\[CrossRef\]](#)
7. Skaalvik, E.M.; Skaalvik, S. Job satisfaction, stress and coping strategies in the teaching profession—What do teachers say? *Int. Educ. Stud.* **2015**, *8*, 181–192. [\[CrossRef\]](#)
8. Agyapong, B.; Obuobi-Donkor, G.; Burbach, L.; Wei, Y. Stress, burnout, anxiety, and depression among teachers: A scoping review. *Int. J. Environ. Res. Public Health* **2022**, *19*, 10706. [\[CrossRef\]](#)
9. Gkintoni, E.; Antonopoulou, H.; Sortwell, A.; Halkiopoulos, C. Challenging Cognitive Load Theory: The Role of Educational Neuroscience and Artificial Intelligence in Redefining Learning Efficacy. *Brain Sci.* **2025**, *15*, 203. [\[CrossRef\]](#)
10. Cordes, C.L.; Dougherty, T.W. A review and an integration of research on job burnout. *Acad. Manag. Rev.* **1993**, *18*, 621–656. [\[CrossRef\]](#)
11. Harmsen, R.; Helms-Lorenz, M.; Maulana, R.; van Veen, K. The relationship between beginning teachers' stress causes, stress responses, teaching behaviour and attrition. *Teach. Teach.* **2018**, *24*, 626–643. [\[CrossRef\]](#)
12. Gkintoni, E.; Dimakos, I.; Nikolaou, G. Cognitive Insights from Emotional Intelligence: A Systematic Review of EI Models in Educational Achievement. *Emerg. Sci. J.* **2025**, *8*, 262–297. [\[CrossRef\]](#)
13. Klusmann, U.; Richter, D.; Lüdtke, O. Teachers' emotional exhaustion is negatively related to students' achievement: Evidence from a large-scale assessment study. *J. Educ. Psychol.* **2016**, *108*, 1193–1203. [\[CrossRef\]](#)
14. O'Brien, G.E. *Workplace Stress and Coping Among Teachers*; Australian Council for Educational Research: Camberwell, Australia, 2007.
15. Ransford, C.R.; Greenberg, M.T.; Domitrovich, C.E.; Small, M.; Jacobson, L.; Sanetti, L.; Kratochwill, T. The role of teachers' psychological experiences and perceptions of curriculum supports on the implementation of a social and emotional learning curriculum. *Sch. Psychol. Rev.* **2009**, *38*, 510–532. [\[CrossRef\]](#)
16. Taris, T.W.; Leisink, P.L.; Schaufeli, W.B. Applying occupational health theories to educator stress: Contributions of the Job Demands-Resources model. In *Educator Stress: An Occupational Health Perspective*; McIntyre, T.M., McIntyre, S.E., Francis, D.J., Eds.; Springer: Berlin/Heidelberg, Germany, 2017; pp. 237–260.
17. Chang, M.-L. An appraisal perspective of teacher burnout: Examining the emotional work of teachers. *Educ. Psychol. Rev.* **2009**, *21*, 193–218. [\[CrossRef\]](#)
18. Carroll, A.; Forrest, K.; Sanders-O'Connor, E.; Flynn, L.; Bower, J.M.; Fynes-Clinton, S.; York, A.; Ziaei, M. Teacher stress and burnout in Australia: Examining the role of intrapersonal and environmental factors. *Soc. Psychol. Educ.* **2022**, *25*, 441–469. [\[CrossRef\]](#) [\[PubMed\]](#)
19. Gkintoni, E.; Halkiopoulos, C.; Antonopoulou, H. Neuroleadership as an Asset in Educational Settings: An Overview. *Emerg. Sci. J.* **2022**, *6*, 893–904. [\[CrossRef\]](#)
20. Petropoulou, A.; Antonopoulou, H.; Vlachou, A.A.; Gkintoni, E.; Halkiopoulos, C. Social-Cognitive Factors in Antisocial Behavior and School Violence: A Cross-Sectional Analysis of Greek Vocational Students. *Children* **2025**, *12*, 1647. [\[CrossRef\]](#)

21. El Helou, M.; Nabhani, M.; Bahous, R. Teachers' views on causes leading to their burnout. *Sch. Leadersh. Manag.* **2016**, *36*, 551–567. [\[CrossRef\]](#)
22. Richards, J. Teacher stress and coping strategies: A national snapshot. *Educ. Forum* **2012**, *76*, 299–316. [\[CrossRef\]](#)
23. Boles, J.S.; Dean, D.H.; Ricks, J.M.; Short, J.C.; Wang, G. The dimensionality of the Maslach Burnout Inventory across small business owners and educators. *J. Vocat. Behav.* **2000**, *56*, 12–34. [\[CrossRef\]](#)
24. Schaufeli, W.B.; Bakker, A.B.; Salanova, M. The measurement of work engagement with a short questionnaire: A cross-national study. *Educ. Psychol. Meas.* **2006**, *66*, 701–716. [\[CrossRef\]](#)
25. Brenner, S.O. *Work, Health, and Well-Being*; Arbetarskyddsfonten: Stockholm, Sweden, 1982.
26. Rubilar, N.V.; Oros, L.B. Stress and burnout in teachers during times of pandemic. *Front. Psychol.* **2021**, *12*, 756007. [\[CrossRef\]](#) [\[PubMed\]](#)
27. Räsänen, K.; Pietarinen, J.; Soini, T.; Väisänen, P.; Pyhältö, K. Experienced risk of burnout among teachers with persistent turnover intentions. *Teach. Dev.* **2022**, *26*, 317–337. [\[CrossRef\]](#)
28. Madigan, D.J.; Kim, L.E.; Glandorf, H.L.; Kavanagh, O. Teacher burnout and physical health: A systematic review. *Int. J. Educ. Res.* **2023**, *119*, 102173. [\[CrossRef\]](#)
29. Mijakoski, D.; Cheptea, D.; Marca, S.C.; Shoman, Y.; Caglayan, C.; Bugge, M.D.; Gnesi, M.; Godderis, L.; Kiran, S.; McElvenny, D.M.; et al. Determinants of burnout among teachers: A systematic review of longitudinal studies. *Int. J. Environ. Res. Public Health* **2022**, *19*, 5776. [\[CrossRef\]](#)
30. Persson, R.; Leo, U.; Arvidsson, I.; Håkansson, C.; Nilsson, K.; Österberg, K. Prevalence of exhaustion symptoms and associations with school level, length of work experience and gender: A nationwide cross-sectional study of Swedish principals. *BMC Public Health* **2021**, *21*, 331. [\[CrossRef\]](#)
31. Berjot, S.; Weber, T.; Huyghebaert-Zouaghi, T. Burnout integrative measure: A preliminary validation among French college students. *Front. Psychol.* **2022**, *13*, 904367. [\[CrossRef\]](#)
32. Johnson, M.W. Use of the Copenhagen Burnout Inventory among U.S. academic librarians: Examining construct validity through factor structure and model fit. *J. Acad. Libr.* **2024**, *50*, 102922. [\[CrossRef\]](#)
33. Delladio, S.; Caputo, A. When the tank is empty: Reviewing burnout in entrepreneurship. *J. Small Bus. Manag.* **2025**, *63*, 2320–2356. [\[CrossRef\]](#)
34. Nadon, L.; De Beer, L.T.; Morin, A.J.S. Should burnout be conceptualized as a mental disorder? *Behav. Sci.* **2022**, *12*, 82. [\[CrossRef\]](#)
35. Bakker, A.B.; Demerouti, E. The Job Demands-Resources model: State of the art. *J. Manag. Psychol.* **2007**, *22*, 309–328. [\[CrossRef\]](#)
36. McCarthy, C.J.; Lambert, R.G.; Lineback, S.; Fitchett, P.; Baddouh, P.G. Assessing teacher appraisals and stress in the classroom: Review of the classroom appraisal of resources and demands. *Educ. Psychol. Rev.* **2015**, *28*, 577–603. [\[CrossRef\]](#)
37. Demerouti, E.; Bakker, A.B.; Nachreiner, F.; Schaufeli, W.B. The job demands-resources model of burnout. *J. Appl. Psychol.* **2001**, *86*, 499–512. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Bakker, A.B.; de Vries, J.D. Job Demands–Resources theory and self-regulation: New explanations and remedies for job burnout. *Anxiety Stress Coping* **2020**, *34*, 1–21. [\[CrossRef\]](#)
39. Demerouti, E.; Bakker, A.B. Job Demands-Resources theory in times of crises: New propositions. *Organ. Psychol. Rev.* **2022**, *13*, 209–236. [\[CrossRef\]](#)
40. Maslach, C.; Leiter, M.P. *The Burnout Challenge: Managing People's Relationships with Their Jobs*; Harvard University Press: Cambridge, MA, USA, 2022. [\[CrossRef\]](#)
41. Bakker, A.B.; Demerouti, E.; Sanz-Vergel, A. Job Demands-Resources theory: Ten years later. *Annu. Rev. Organ. Psychol. Organ. Behav.* **2023**, *10*, 25–53. [\[CrossRef\]](#)
42. Bakker, A.B.; Schaufeli, W.B.; Leiter, M.P.; Taris, T.W. Work engagement: An emerging concept in occupational health psychology. *Work Stress* **2008**, *22*, 187–200. [\[CrossRef\]](#)
43. Edú-Valsania, S.; Laguía, A.; Moriano, J.A. Burnout: A review of theory and measurement. *Int. J. Environ. Res. Public Health* **2022**, *19*, 1780. [\[CrossRef\]](#)
44. Reed, B.N. Turning the tide: Addressing threats to pharmacist well-being through work redesign. *J. Am. Coll. Clin. Pharm.* **2023**, *6*, 742–750. [\[CrossRef\]](#)
45. Alarcon, G.M. A meta-analysis of burnout with job demands, resources, and attitudes. *J. Vocat. Behav.* **2011**, *79*, 549–562. [\[CrossRef\]](#)
46. Leclercq, C.; Hansez, I. Temporal stages of burnout: How to design prevention? *Int. J. Environ. Res. Public Health* **2024**, *21*, 1617. [\[CrossRef\]](#)
47. Fernández-Suárez, I.; García-González, M.A.; Torrano, F.; García-González, G. Study of the prevalence of burnout in university professors in the period 2005–2020. *Educ. Res. Int.* **2021**, *2021*, 7810659. [\[CrossRef\]](#)
48. Holmes, E.A.; O'Connor, R.C.; Perry, V.H.; Tracey, I.; Wessely, S.; Arseneault, L.; Ballard, C.; Christensen, H.; Silver, R.C.; Everall, I.; et al. Multidisciplinary research priorities for the COVID-19 pandemic: A call for action for mental health science. *Lancet Psychiatry* **2020**, *7*, 547–560. [\[CrossRef\]](#) [\[PubMed\]](#)

49. Sokal, L.; Trudel, L.E.; Babb, J. Canadian teachers' attitudes toward change, efficacy, and burnout during the COVID-19 pandemic. *Int. J. Educ. Res. Open* **2020**, *1*, 100016. [\[CrossRef\]](#)
50. Broadbent, J.; Ajjawi, R.; Bearman, M.; Boud, D.; Dawson, P. Beyond emergency remote teaching: Did the pandemic lead to lasting change in university courses? *Int. J. Educ. Technol. High. Educ.* **2023**, *20*, 58. [\[CrossRef\]](#)
51. Srivastava, P.; Sehgal, T.; Jain, R.; Kaur, P.; Luukela-Tandon, A. Knowledge management during emergency remote teaching: An interpretative phenomenological analysis of the transition experiences of faculty members. *J. Knowl. Manag.* **2024**, *28*, 78–105. [\[CrossRef\]](#)
52. González-García, H.; Martinent, G.; Morales, A.T. Perceived Coach Leadership Profiles and Relationship with Burnout, Coping, and Emotions. *Front. Psychol.* **2019**, *10*, 1785. [\[CrossRef\]](#)
53. Antonopoulou, H.; Halkiopoulou, C.; Barlou, O.; Beligiannis, G.N. Associations between Traditional and Digital Leadership in Academic Environment: During the COVID-19 Pandemic. *Emerg. Sci. J.* **2021**, *5*, 405–428. [\[CrossRef\]](#)
54. Bruria, A.; Maya, S.-T.; Gadi, S.; Orna, T. Impact of emergency situations on resilience at work and burnout of hospital's healthcare personnel. *Int. J. Disaster Risk Reduct.* **2022**, *76*, 102994. [\[CrossRef\]](#)
55. Wittmers, A.; Maier, G.W. Leaders' mental health in times of crisis: Work intensification, emotional demands and the moderating role of organizational support and self-efficacy. *Front. Psychol.* **2023**, *14*, 1122881. [\[CrossRef\]](#)
56. Farber, B.A. Stress and burnout in suburban teachers. *J. Educ. Res.* **1984**, *77*, 325–331. [\[CrossRef\]](#)
57. Van Dick, R.; Wagner, U. Stress and strain in teaching: A structural equation approach. *Br. J. Educ. Psychol.* **2001**, *71*, 243–259. [\[CrossRef\]](#) [\[PubMed\]](#)
58. Antoniou, A.; Polychroni, F.; Vlachakis, A. Gender and age differences in occupational stress and professional burnout between primary and high-school teachers in Greece. *J. Manag. Psychol.* **2006**, *21*, 682–690. [\[CrossRef\]](#)
59. Bilge, F. Examining the burnout of academics in relation to job satisfaction and other factors. *Soc. Behav. Pers. Int. J.* **2006**, *34*, 1151–1160. [\[CrossRef\]](#)
60. Burke, R.J.; Greenglass, E. Work stress, role conflict, social support, and psychological burnout among teachers. *Psychol. Rep.* **1993**, *73*, 371–380. [\[CrossRef\]](#)
61. Ghorpade, J.; Lackritz, J.; Singh, G. Burnout and personality: Evidence from academia. *J. Career Assess.* **2007**, *15*, 240–256. [\[CrossRef\]](#)
62. Sánchez-Pujalte, L.; Mateu, D.N.; Etchezahar, E.; Yepes, T.G. Teachers' burnout during COVID-19 pandemic in Spain: Trait emotional intelligence and socioemotional competencies. *Sustainability* **2021**, *13*, 7259. [\[CrossRef\]](#)
63. Nagle, E.; Giskevica, I.; Rajevska, O.; Ivanovs, A.; Mihailova, S.; Skruzkalne, I. Factors affecting healthcare workers burnout and their conceptual models: Scoping review. *BMC Psychol.* **2024**, *12*, 637. [\[CrossRef\]](#)
64. Le, T.V.; Barboza-Wilkes, C. How self-determination moderates burnout in public organizations across gender, race, and generation. *Public Pers. Manag.* **2024**, *53*, 113–147. [\[CrossRef\]](#)
65. Chmiel, J.; Wiażewicz-Wójtowicz, P.; Stepień-Słodkowska, M. Neural correlates of burnout syndrome based on electroencephalography (EEG)—A mechanistic review and discussion of burnout syndrome cognitive aspects. *J. Clin. Med.* **2025**, *14*, 1895. [\[CrossRef\]](#)
66. Ansley, B.M.; Houchins, D.E.; Varjas, K.; Roach, A.; Patterson, D.; Hendrick, R. The impact of an online stress intervention on burnout and teacher efficacy. *Teach. Teach. Educ.* **2021**, *98*, 103251. [\[CrossRef\]](#)
67. Byrne, B.M. Burnout: Investigating the impact of background variables for elementary, intermediate, secondary, and university educators. *Teach. Teach. Educ.* **1991**, *7*, 197–209. [\[CrossRef\]](#)
68. Kantas, A.; Vassilaki, E. Burnout in Greek teachers: Main findings and validity of the Maslach Burnout Inventory. *Work. Stress* **1997**, *11*, 94–100. [\[CrossRef\]](#)
69. Koehler, M.J.; Mishra, P. Introducing TPCK. In *Handbook of Technological Pedagogical Content Knowledge (TPCK) for Educators*; AACTE Committee on Innovation and Technology, Ed.; Routledge: London, UK, 2008; pp. 3–29.
70. Maslach, C.; Schaufeli, W.B.; Leiter, M.P. Job burnout. *Annu. Rev. Psychol.* **2001**, *52*, 397–422. [\[CrossRef\]](#) [\[PubMed\]](#)
71. Vassilopoulos, S.P. Job burnout and work engagement in primary school teachers: The role of the school context. *Hell. J. Psychol.* **2012**, *9*, 18–39.
72. Söderholm, F.; Viljaranta, J.; Hirvonen, R.; Tuominen, H.; Lappalainen, K.; Holopainen, L. The development of exhaustion, cynicism, and inadequacy in general upper secondary education: Perceived and received support, student engagement, and gender as predictors. *Scand. J. Educ. Res.* **2025**, *69*, 1271–1287. [\[CrossRef\]](#)
73. Tikkanen, L.; Haverinen, K.; Pyhältö, K.; Pietarinen, J.; Soini, T. Differences in teacher burnout between schools: Exploring the effect of proactive strategies on burnout trajectories. *Front. Educ.* **2022**, *7*, 858896. [\[CrossRef\]](#)
74. Boone, A.; Elst, T.V.; Vandenbroeck, S.; Godderis, L. Burnout profiles among young researchers: A latent profile analysis. *Front. Psychol.* **2022**, *13*, 839728. [\[CrossRef\]](#)
75. Ghasemi, F. The effects of dysfunctional workplace behavior on teacher emotional exhaustion: A moderated mediation model of perceived social support and anxiety. *Psychol. Rep.* **2024**, *127*, 2526–2551. [\[CrossRef\]](#)

76. Kokkinos, C.M.; Davazoglou, A.M. Special education teachers under stress: Evidence from a Greek national study. *Educ. Psychol.* **2009**, *29*, 407–424. [[CrossRef](#)]
77. Kokkinos, C.M. Job stressors, personality and burnout in primary school teachers. *Br. J. Educ. Psychol.* **2007**, *77*, 229–243. [[CrossRef](#)] [[PubMed](#)]
78. Turner, M.; Holdsworth, S. Developing resilience: Examining the protective factors of early career construction professionals. *Constr. Manag. Econ.* **2023**, *41*, 805–819. [[CrossRef](#)]
79. Guo, L. Unsettled horizon: Adolescents' career expectations in the volatile, uncertain, complex, and ambiguous contexts. *Humanit. Soc. Sci. Commun.* **2025**, *12*, 5298. [[CrossRef](#)]
80. Frolova, Y. Navigating career uncertainty: The role of proactive decision-making, sense of emptiness and constructivist beliefs in professional fulfillment. *Int. J. Organ. Theory Behav.* **2025**. [[CrossRef](#)]
81. Tajeddin, Z.; Keshvari, Z. Emotional tensions in identity transition from teachers to teacher educators: Voices from early-career teacher educators. *Asian-Pac. J. Second. Foreign Lang. Educ.* **2025**, *10*, 14. [[CrossRef](#)]
82. Rohwedder, A.-B.; Le Fevre, D.M. Effective early career VET teaching: The importance of developing adaptive expertise in workplace learning. *Int. J. Train. Res.* **2025**, *23*, 1–14. [[CrossRef](#)]
83. Bai, Q.; Nam, B.H.; English, A.S.; Marshall, R.C.; Tian, X. Mental health and early career adaptation challenges of Chinese returnees and Korean ambitionists among the STEAM doctoral cohorts in the post-pandemic academic job market. *Comp. A J. Comp. Int. Educ.* **2025**, *55*, 1030–1050. [[CrossRef](#)]
84. Evers, W.J.G.; Brouwers, A.; Tomic, W. Burnout and self-efficacy: A study on teachers' beliefs when implementing an innovative educational system in the Netherlands. *Br. J. Educ. Psychol.* **2002**, *72*, 227–243. [[CrossRef](#)]
85. Pereira, H.; Gonçalves, V.O.; de Assis, R.M. Burnout, organizational self-efficacy and self-esteem among Brazilian teachers during the COVID-19 pandemic. *Eur. J. Investig. Health Psychol. Educ.* **2021**, *11*, 795–803. [[CrossRef](#)]
86. Skaalvik, E.M.; Skaalvik, S. Dimensions of teacher self-efficacy and relations with strain factors, perceived collective teacher efficacy, and teacher burnout. *J. Educ. Psychol.* **2007**, *99*, 611–625. [[CrossRef](#)]
87. Vladut, C.I.; Kállay, É. Psycho-emotional and organizational aspects of burnout in a sample of Romanian teachers. *Cogn. Brain Behav.* **2011**, *14*, 331–358.
88. Cao, Y.; Gao, L.; Fan, L.; Jiao, M.; Li, Y.; Ma, Y. The influence of emotional intelligence on job burnout of healthcare workers and mediating role of workplace violence: A cross-sectional study. *Front. Public Health* **2022**, *10*, 892421. [[CrossRef](#)] [[PubMed](#)]
89. Almeneessier, A.S.; Azer, S.A. Exploring the relationship between burnout and emotional intelligence among academics and clinicians at King Saud University. *BMC Med. Educ.* **2023**, *23*, 689. [[CrossRef](#)] [[PubMed](#)]
90. Bagatini, A.; Bumaguin, D.B.; Bagatini, L.Z.; Ardais, A.P. The relationship between burnout syndrome and emotional intelligence in anesthesiologists during the COVID-19 pandemic: A cross-sectional study. *J. Patient Saf. Risk Manag.* **2025**, *30*, 149–156. [[CrossRef](#)]
91. Merino-Soto, C.; Angulo-Ramos, M.; Llaja-Rojas, V.; Chans, G.M. Academic performance, emotional intelligence, and academic burnout: A cross-sectional study of a mediational effect in nursing students. *Nurse Educ. Today* **2024**, *139*, 106221. [[CrossRef](#)]
92. Pietarinen, J.; Pyhältö, K.; Haverinen, K.; Leskinen, E.; Soini, T. Is individual- and school-level teacher burnout reduced by proactive strategies? *Int. J. Sch. Educ. Psychol.* **2021**, *9*, 340–355. [[CrossRef](#)]
93. Ahmed, A.K.; Atta, M.H.R.; El-Monshed, A.H.; Mohamed, A.I. The effect of toxic leadership on workplace deviance: The mediating effect of emotional exhaustion, and the moderating effect of organizational cynicism. *BMC Nurs.* **2024**, *23*, 308. [[CrossRef](#)]
94. Braun, S.S.; Schonert-Reichl, K.A.; Roeser, R.W. Effects of teachers' emotion regulation, burnout, and life satisfaction on student well-being. *J. Appl. Dev. Psychol.* **2020**, *69*, 101151. [[CrossRef](#)]
95. Agina, M.F.; Khairy, H.A.; Fatah, M.A.A.; Manaa, Y.H.; Abdallah, R.M.; Aliane, N.; Afaneh, J.; Al-Romeedy, B.S. Distributive injustice and work disengagement in the tourism and hospitality industry: Mediating roles of workplace negative gossip and organizational cynicism. *Sustainability* **2023**, *15*, 15011. [[CrossRef](#)]
96. Bibou-Nakou, I.; Stogiannidou, A.; Kiosseoglou, G. The relation between teacher burnout and teachers' attributions and practices regarding school behaviour problems. *Sch. Psychol. Int.* **1999**, *20*, 209–217. [[CrossRef](#)]
97. Vandenberghe, R.; Huberman, A.M. (Eds.) *Understanding and Preventing Teacher Burnout: A Sourcebook of International Research and Practice*; Cambridge University Press: Cambridge, UK, 1999.
98. Papadakis, N.; Drakaki, M.; Saridaki, S.; Amanaki, E.; Dimari, G. Educational capital/level and its association with precarious work and social vulnerability among youth, in EU and Greece. *Int. J. Educ. Res.* **2022**, *112*, 101921. [[CrossRef](#)]
99. Koutra, K.; Mavroeides, G.; Triliva, S. Of caring: Prevalence and associated factors of burnout, compassion fatigue, and compassion satisfaction among mental health professionals in Greece. *Psychol. Int.* **2025**, *7*, 58. [[CrossRef](#)]
100. Pakou, V.; Andrianopoulos, I.; Gouva, M.; Dragioti, E.; Tatsis, F.; Papathanakos, G.; Ntantana, A.; Antoniadou, E.; Fligou, F.; Papageorgiou, E.; et al. Burnout and mental health symptoms among ICU professionals in Greece: A national comparative study between the COVID-19 pandemic and pre-pandemic eras. *Cureus* **2025**, *17*, e84555. [[CrossRef](#)] [[PubMed](#)]

101. Katsarou, E.; Chatzipanagiotou, P.; Sougari, A.-M. A systematic review on teachers' well-being in the COVID-19 era. *Educ. Sci.* **2023**, *13*, 927. [\[CrossRef\]](#)
102. Kim, L.E.; Asbury, K. 'Like a rug had been pulled from under you': The impact of COVID-19 on teachers in England during the first six weeks of the UK lockdown. *Br. J. Educ. Psychol.* **2020**, *90*, 1062–1083. [\[CrossRef\]](#)
103. Pressley, T. Factors contributing to teacher burnout during COVID-19. *Educ. Res.* **2021**, *50*, 325–327. [\[CrossRef\]](#)
104. Oliveira, S.; Roberto, M.S.; Pereira, N.S.; Marques-Pinto, A.; Veiga-Simão, A.M. Impacts of Social and Emotional Learning Interventions for Teachers on Teachers' Outcomes: A Systematic Review with Meta-Analysis. *Front. Psychol.* **2021**, *12*, 677217. [\[CrossRef\]](#)
105. Malinauskas, R.; Malinauskiene, V. Characteristics of stress and burnout among Lithuanian university coaches: A pre-pandemic coronavirus and post-pandemic period comparison. *Healthcare* **2023**, *11*, 2096. [\[CrossRef\]](#)
106. Sikaras, C.; Pachi, A.; Alikanioti, S.; Ilias, I.; Sideri, E.P.; Tselebis, A.; Panagiotou, A. Occupational burnout and insomnia in relation to psychological resilience among Greek nurses in the post-pandemic era. *Behav. Sci.* **2025**, *15*, 126. [\[CrossRef\]](#)
107. Katsogiannis, I.; Manara, E.; Peletidi, A.; Bistaraki, A.; Constantinides, T.; Kontogiorgis, C. Occupational burnout and job satisfaction among community pharmacists. *Explor. Res. Clin. Soc. Pharm.* **2024**, *14*, 100445. [\[CrossRef\]](#)
108. Patra, D.; Bharti, N.; Dutta, M. Are better equipped schools delivering better student outcomes? Evidence from India. *Indian J. Hum. Dev.* **2025**, *19*, 09737030251346751. [\[CrossRef\]](#)
109. Giannopoulos, A.; Caouris, Y.G.; Souliotis, M.; Santamouris, M. Characteristics of the urban heat island effect, in the coastal city of Patras, Greece. *Int. J. Sustain. Energy* **2022**, *41*, 556–571. [\[CrossRef\]](#)
110. Petridou, V. The Organization for Economic Co-operation and Development (OECD) experts' contribution in the establishment of the University of Patras. *Plan. Perspect.* **2022**, *37*, 1029–1049. [\[CrossRef\]](#)
111. Antonopoulou, H.; Halkiopoulos, C.; Gkintoni, E.; Katsimpelis, A. Application of Gamification Tools for Identification of Neurocognitive and Social Function in Distance Learning Education. *Int. J. Learn. Teach. Educ. Res.* **2022**, *21*, 367–400. [\[CrossRef\]](#)
112. Grządzielewska, M. Using machine learning in burnout prediction: A survey. *Child Adolesc. Soc. Work. J.* **2021**, *38*, 175–180. [\[CrossRef\]](#)
113. Johnson, T.; Shamroukh, S. Predictive modeling of burnout based on organizational culture perceptions among health systems employees: A comparative study using correlation, decision tree, and regression models. *Sci. Rep.* **2024**, *14*, 56771. [\[CrossRef\]](#)
114. Merhbene, G.; Nath, S.; Puttick, A.R.; Kurpicz-Briki, M. BurnoutEnsemble: Augmented intelligence to detect indications for burnout in clinical psychology. *Front. Big Data* **2022**, *5*, 863100. [\[CrossRef\]](#)
115. Alexaki, P.-S.; Antonopoulou, H.; Gkintoni, E.; Adamopoulos, N.; Halkiopoulos, C. Psychological dimensions of professional burnout in special education: A cross-sectional behavioral data analysis of emotional exhaustion, personal achievement, and depersonalization. *Int. J. Environ. Res. Public Health* **2025**, *22*, 1420. [\[CrossRef\]](#)
116. Regli, A.; Antonopoulou, H.; Beligiannis, G.N.; Asimakopoulos, G.; Halkiopoulos, C. Beyond Digital Natives: A System-Level Analysis of Institutional Barriers and Teacher Experience in Secondary School ICT Integration. *Sustainability* **2026**, *18*, 1108. [\[CrossRef\]](#)
117. Zabel, R.H.; Zabel, M.K. Revisiting burnout among special education teachers: Do age, experience, and preparation still matter? *Teach. Educ. Spec. Educ.* **2001**, *24*, 128–139. [\[CrossRef\]](#)
118. Han, H. Fuzzy clustering algorithm for university students' psychological fitness and performance detection. *Heliyon* **2023**, *9*, e18550. [\[CrossRef\]](#)

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